

TASNIM UL ISLAM

B.Sc in Bioinformatics Engineering · Bangladesh Agricultural University

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PROFILE

An experienced graduate researcher in bioinformatics, with expertise in fish genomics, population genetics, mitochondrial genomics, and deep learning for biological sequence analysis along with software and pipeline development for bioinformatics tools. I have been involved in the reference-guided genome assembly, mitogenomics, phylogenetics, and population-genetic analysis of threatened and commercially important fishes at the Bangladesh Fisheries Research Institute. My undergraduate research is dedicated to the identification of plastic-degrading genes through the use of DNABERT-2 and ESM-2 for DNA–protein cross-modal learning. Author of a Springer paper that has been peer-reviewed and of several manuscripts that are either in the process of being reviewed or prepared. Seeking M.S./Ph.D. opportunities in conservation genomics, bioinformatics, and computational biology.

EDUCATION

B.Sc. in Bioinformatics Engineering 2020 – Present
Bangladesh Agricultural University, Mymensingh
Session 2020–21
CGPA 3.463 / 4.00 | Rank: 5th
Expected graduation: 2026

HSC Kushtia Govt. College (2020)
GPA 5.00/5.00

SSC Kushtia Zilla School (2018)
GPA 5.00/5.00

UNDERGRADUATE RESEARCH / THESIS

A DNA-Protein Cross Modal Deep Learning Framework for Discovering Plastic Degrading Genes in Bacteria with In Silico Analysis (Equal contribution; second author) 2025–2026

- Curated a cluster-aware CDS–protein dataset of 1,501 sequences using CD-HIT to minimize homology leakage.
- Evaluated NT-v2, HyenaDNA, ProtT5, Ankh, DNABERT-2 and ESM-2; selected partially fine-tuned DNABERT-2 + ESM-2
- Projected DNA and protein representations into a shared 512-dimensional latent space using InfoNCE and implemented 8-head bidirectional cross-attention for multimodal fusion.
- Designed the model so protein information guides training while DNA sequence alone is sufficient during inference
- Achieved 0.910 accuracy, 0.872 F1, 0.968 ROC-AUC, 0.958 PR-AUC and 0.806 MCC on the held-out test set

RESEARCH EXPERIENCE

Research Intern — Bioinformatics 2026

Fish Genomics Laboratory, Bangladesh Fisheries Research Institute (BFRI)

- Extracted genomic DNA from fish tissue, PCR-amplified mitochondrial targets, and confirmed products by agarose gel electrophoresis before Illumina short-read sequencing.
- Assembled a reference-guided consensus genome of *Tor tor* from paired Illumina libraries (FastQC, fastp, BWA-MEM, SAMtools, BCFtools) and assessed completeness with BUSCO; annotated loci relevant to growth, immunity, and environmental tolerance.
- Assembled and annotated complete mitochondrial genomes (~16 kb) of *T. tor* and *Tenualosa ilisha* (MitoFish/MITOS2); wrote Python/R scripts for RSCU, Ka/Ks, and nucleotide-composition asymmetry.
- Analysed 900+ partial COI and Cyt *b* sequences across sampling sites (PhyloSuite, MAFFT, IQ-TREE); estimated population structure with AMOVA, pairwise FST, PCoA, and neutrality tests (Tajima's D, Fu's Fs), and mapped localities in Python (Cartopy).
- Annotated hypothetical bacterial proteins (AutoDock, PyRx, PyMOL, BIOVIA Discovery Studio). Contributed computational analyses to the fisheries manuscripts currently under review, revision or preparation.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed

- **UI Islam, T.**, Shahriar, S., Uddin, M., Hassan, M. R. (2025). Pseudo-Knighted Cocktail Shaker Sort. In *Proceedings of Trends in Electronics and Health Informatics (TEHI 2023)*, Lecture Notes in Networks and Systems, vol. 1034, pp. 149–167. Springer, Singapore. [doi:10.1007/978-981-97-3937-0_11](https://doi.org/10.1007/978-981-97-3937-0_11) .

Under review and in revision

- Haque, M. A., **UI Islam, T.**, et al. Mitogenomics of Hilsa Shad (*Tenualosa ilisha*): genome architecture, evolutionary patterns, and phylogeny - Manuscript under review.
- Haque, M. A., **UI Islam, T.**, et al. Comparative analysis of mitochondrial markers: genetic diversity, population structure, and demographic history of *Tenualosa ilisha* - Manuscript in revision .

In preparation

- Haque, M. A., et al. (including **UI Islam, T.**). Reference-guided whole-genome analysis of *Tor tor*. Manuscript in preparation.

- Haque, M. A., et al. (including **UI Islam, T.**). Comparative mitochondrial gene analysis of *Tor tor*. Manuscript in preparation.
- **UI Islam, T.**,UI-Arif,M. T., Hossen,M. B.,et al. Genome-Wide Identification and Expression Analysis of the Metacaspase Gene Family in Oryza Species. manuscript in preparation
- Shahriar, S., **UI Islam, T.** (equal contribution), et al. A DNA–protein cross-modal deep-learning framework for discovering plastic-degrading genes in bacteria. Journal version of the undergraduate project; experimental validation planned. Manuscript in preparation.

CONFERENCES & PRESENTATIONS

- BioSeqInsight: An Integrated Local Platform for DNA-to-Protein Sequence Analysis and Structure Prediction. Oral presentation of accepted abstract, Regional Statistical Conference 2026 (BAU DAAS / BSA), April 2026.
- Oral presentation titled “Genome-Wide Identification and Expression Analysis of the Metacaspase Gene Family in Oryza Species.” on agrobiotechnology, genetic engineering and bioinformatics. International Conference on Regenerative Agriculture for Sustainable Food Security, Gazipur Agricultural University, December 2025.
- Contributory poster presentation titled “In-Silico discovery of Sandhoff Disease causing key molecular signatures and therapeutic agents through single-cell RNA-Seq analysis” on International Conference on Applied Statistics and Data Science 2025

SELECTED COMPUTATIONAL PROJECTS

BioSeqInsight 2025 – Present

Python desktop tool for local sequence analysis and structure lookup

- Implemented DNA sequence analysis including GC content, melting temperature, motifs, translation and ORF prediction.
- Integrated ESM Atlas, AlphaFold DB and RCSB PDB for structure retrieval.
- Benchmarked retrieval on 25 public proteins (16/25 live folds; CDK2 pLDDT 90.25). Code: github.com/niloy-999/BioSeqInsight.

scRNA-seq analysis platform 2025 – 2026

Interactive Streamlit app (Python, Scanpy, PyTorch)

- Implements QC, highly variable gene selection, PCA versus VAE embeddings, Leiden/K-means clustering, and marker-gene export.
- Released v0.1.0 with documentation. github.com/niloy-999/scRNA-pipeline · [doi:10.5281/zenodo.22132422](https://doi.org/10.5281/zenodo.22132422).

Also: GWAS coursework presented at Gazipur Agricultural University (Dec 2025); R differential-expression pipeline (2023); other portfolio code at github.com/niloy-999.

TECHNICAL SKILLS

- **Programming:** Python (primary), R, C++, Java, SQL, MATLAB/Octave, LaTeX.
- **Genomics:** FastQC,BWA-MEM, SAMtools,BCFtools, BUSCO,Gemoma,mosdepth,AGAT,QUAST,eggNOG DB,OrthoFinder,CAFE.
- **Population and phylogenetics:** MAFFT, MACSE, IQ-TREE, PhyloSuite, MitoFish/MITOS2, AMOVA, pairwise FST, PCoA, Tajima's D.
- **Structural / functional annotation:** AutoDock, PyRx, PyMOL, BIOVIA Discovery Studio, Cytoscape.
- **Machine learning:** PyTorch, scikit-learn, pandas, NumPy, Scanpy, Streamlit.
- **Deep learning:** DNABERT-2, ESM-2, cross-modal representation learning, attention mechanisms
- **Wet lab:** genomic DNA extraction, PCR, agarose gel electrophoresis.
- **Languages:** Bangla (native), English (fluent).

AWARDS AND RECOGNITION

- Honorable Mention — ICPC Asia Dhaka Regional Contest 2022
- Honorable Mention — ICPC Asia Dhaka Regional Contest 2021
- Best Participant — Virtual Workshop on Bioinformatics: Bridging Science and Computer, BioChemix Club, University of Rajshahi (2024).
- Finalist — Agricultural Olympiad, Season 2; participant, 1st ISCB RSG Bangladesh CompBio Symposium 2023.

LEADERSHIP AND CERTIFICATIONS

- Executive Member, Bioinformatics Engineering Association, BAU (June 2022 – present).
- Applied Data Science Lab — WorldQuant University (2024); Machine Learning — Stanford / Coursera (2022) and Kaggle (2024); Data Analysis with R — Google / Coursera (2024).
- Introduction to Genomic Data Science — Johns Hopkins (2022); Advanced Python for Bioinformatics — Coursera (2021); Java Programming — Duke / Coursera, with honors (2022).

REFERENCES

Mr. Md. Bayazid Hossen(Research Supervisor) Assistant Professor	Md. Amdadul Haque (internship & Research) Scientific Officer,BFRI
Dr. Md. Rakib Hossain(Computational Resource) Professor	Dr. Machbah Uddin (undergraduate project) Professor